

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1110

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 40%

QUESTIONS: Any 4

Total Marks:40

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

*I Tejaswini.M (192411079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Tejaswini.M

ANALYTICAL PROBLEM SOLVING

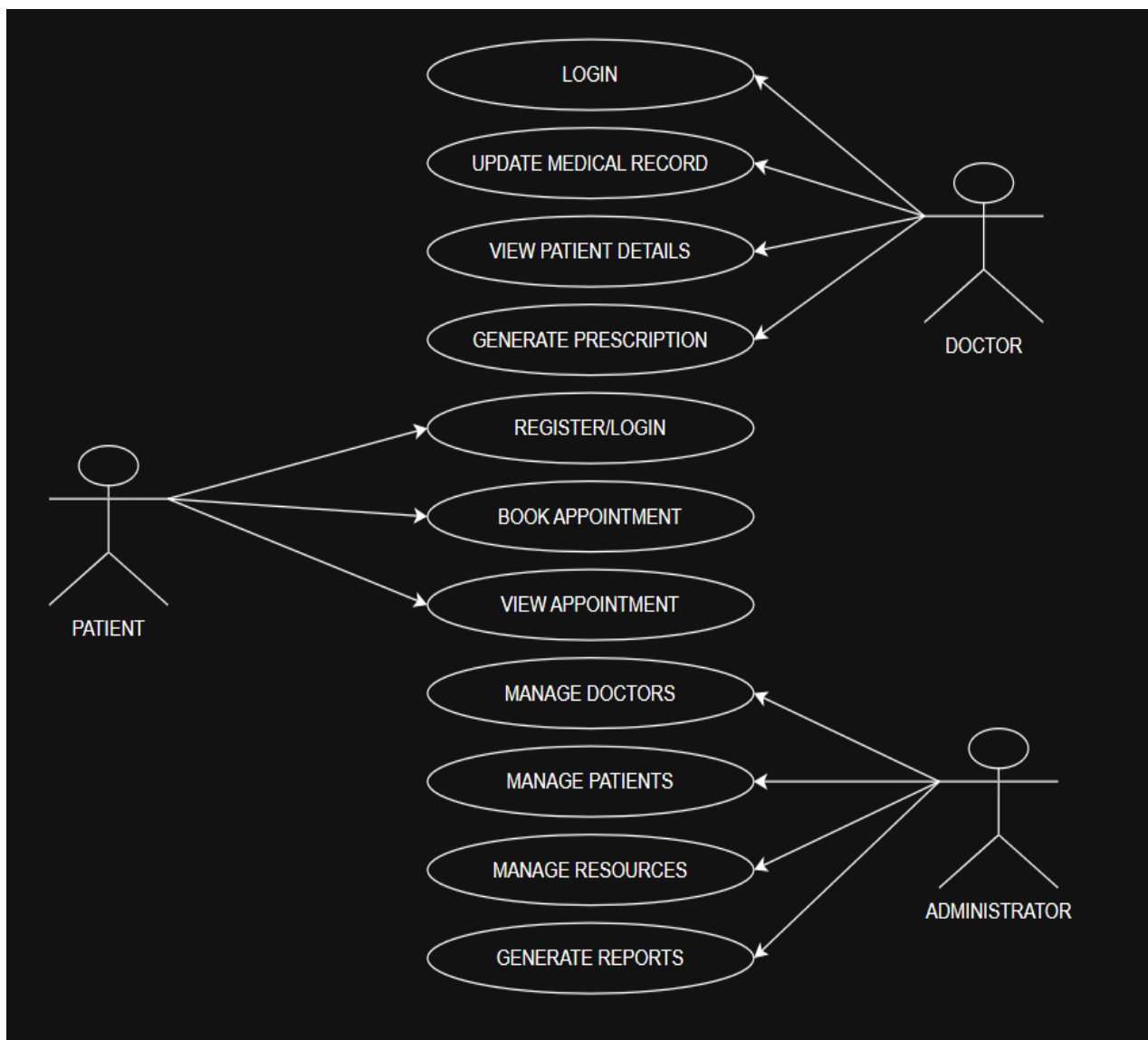
1. Use Case Diagram Problem

A **Hospital Management System** allows patients to book appointments, doctors to update medical records, and administrators to manage hospital resources. The requirements are not clearly documented.

Tasks:

- Identify all actors involved in the system.
- List the primary use cases.
- Define relationships (include, extend, generalization).
- Construct a complete Use Case Diagram representing system functionality.

(10 Marks)



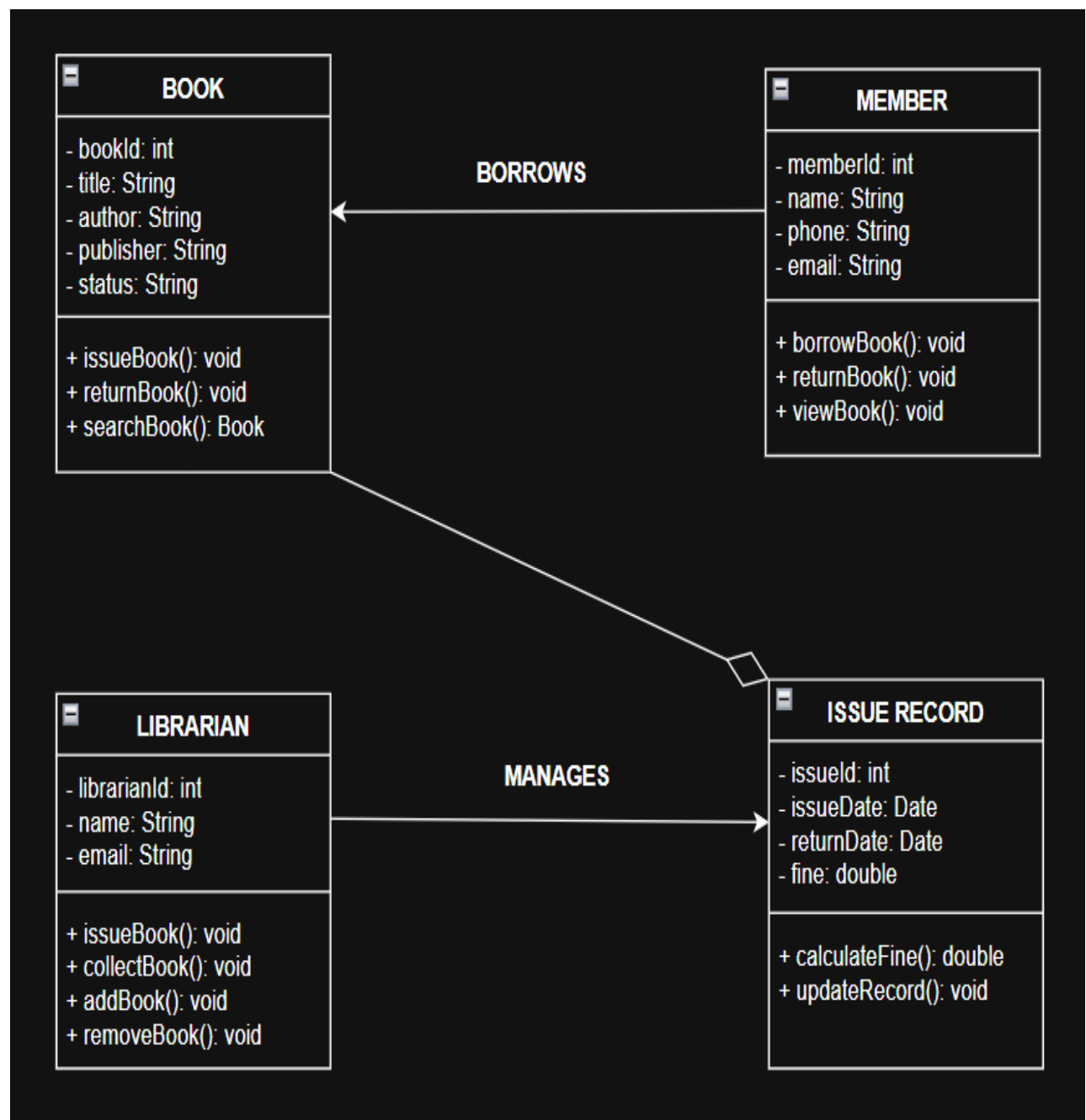
2. Class Diagram Problem

A **Library Management System** contains entities such as Book, Member, Librarian, and IssueRecord. The relationships among entities are not properly defined.

Tasks:

- Identify classes with attributes and methods.
- Define relationships (association, aggregation, inheritance).
- Specify multiplicity between classes.
- Design a detailed Class Diagram for the system.

(10 Marks)



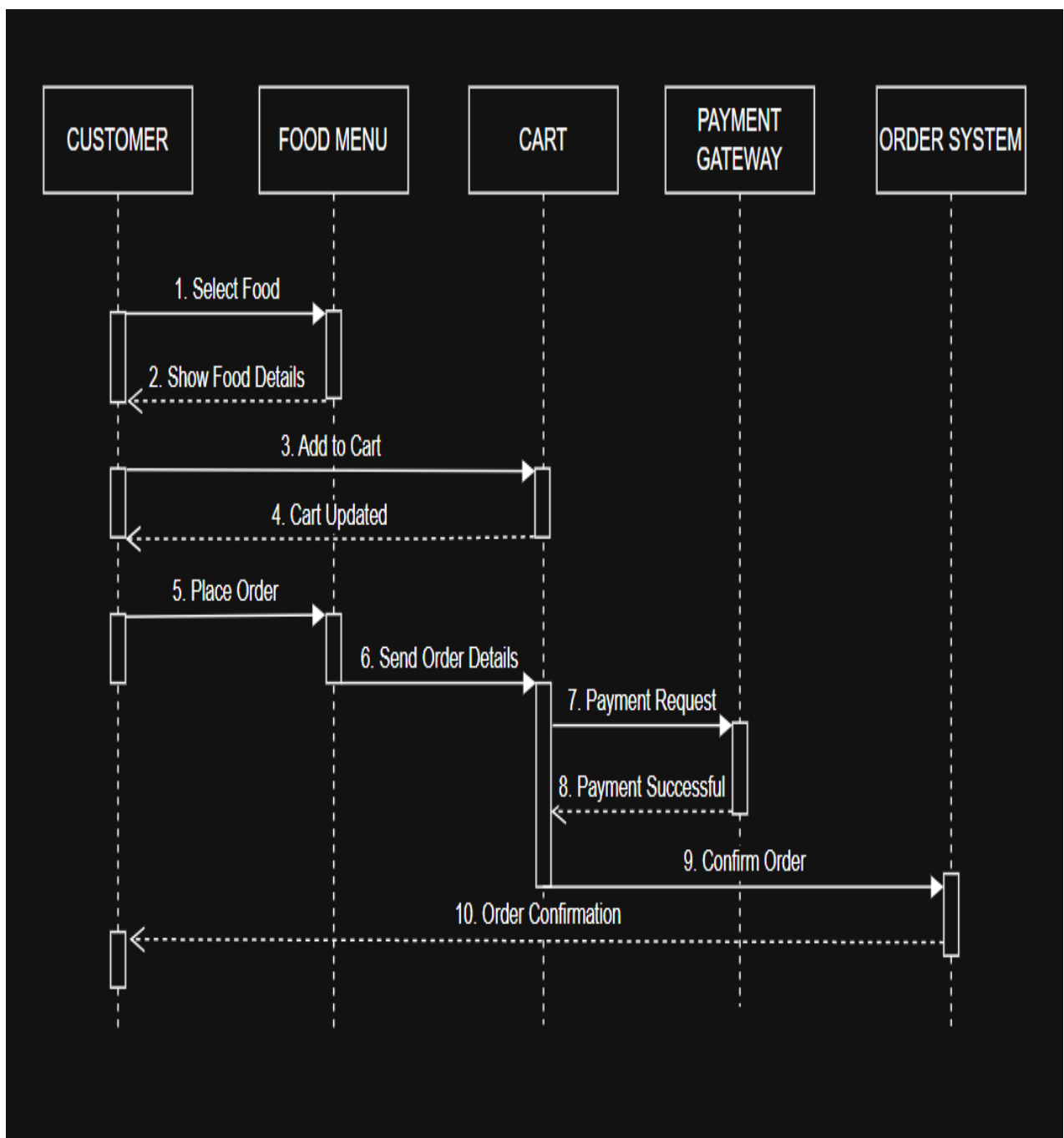
3. Sequence Diagram Problem

In an **Online Food Ordering System**, a customer selects food items, places an order, makes payment, and receives order confirmation. The interaction flow is unclear.

Tasks:

- Identify objects involved in the process.
- Define message flow between objects.
- Represent activation and lifelines.
- Construct a Sequence Diagram showing complete interaction flow.

(10 Marks)



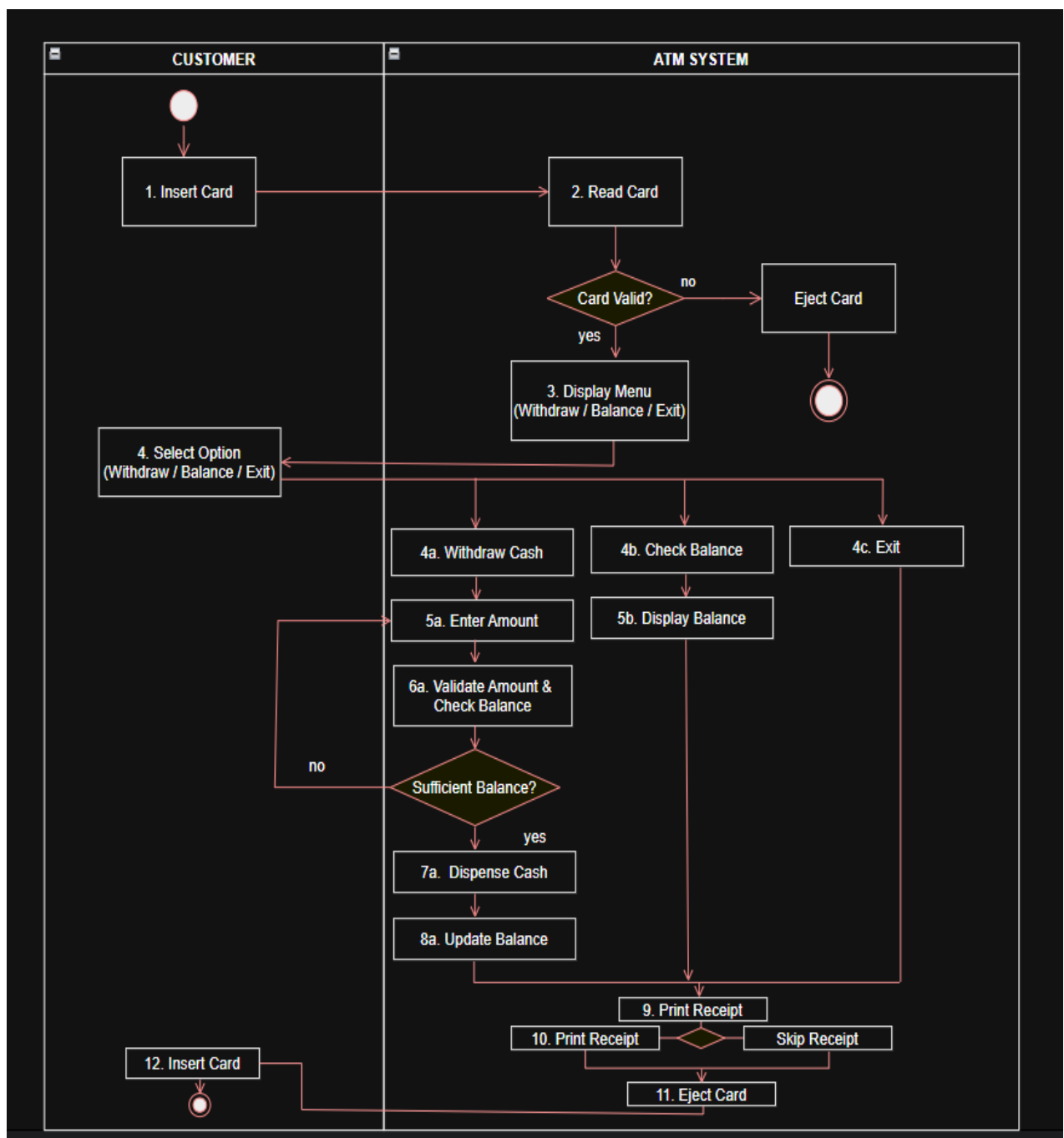
4. Activity Diagram Problem

A **Student Admission System** includes application submission, document verification, fee payment, and enrollment confirmation. The workflow lacks proper decision-making steps.

Tasks:

- Identify all activities in the workflow.
- Add decision nodes for approval/rejection.
- Include parallel processing where applicable.
- Design an optimized Activity Diagram.

(10 Marks)



RUBRICS FOR CALCULATION

ANALYTICAL PROBLEM SOLVING

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation